Turn raw data into answers you can trust. This book teaches data analysis with Python and pandas, one small, clear step at a time, from your first table to complete, tested reports.

Every example is a real program, run on Python 3.12, 3.13, 3.14 and 3.15 before publication, with the exact output printed beneath it. The datasets are generated by a program in the book, so you get exactly the same numbers on your computer.

You'll learn to load spreadsheets, CSV, JSON and databases; clean messy data; filter, group, join and reshape tables; work with dates and text; draw honest charts; test whether a difference is real; write SQL; and check AI-generated analysis code. Then you'll put it all together in six projects: retail sales, a customer survey, an A/B test, a stock study, air quality and health, and a household budget.

Real stories of famous data mistakes show why each habit matters. Every chapter has exercises with tested, worked solutions. Basic Python helps, but no data experience is needed.

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DATA ANALYSIS WITH PYTHON, HANDS-ON

AADHAVAN MUTHURENGAN

sales.py

```
1  import pandas as pd
2
3  sales = pd.DataFrame({
4      "region": ["North", "South", "North"],
5      "amount": [120, 90, 80]})
6  print(sales.groupby("region").sum())
```

```
$ python sales.py
        amount
region
North      200
South       90
```

- CLEAN
- EXPLORE
- GROUP
- CHARTS
- STATISTICS
- SQL

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FROM SPREADSHEETS TO INSIGHT

Clean, Explore, Chart and Report on Real Data, with Every Example Tested

100+ TESTED EXAMPLES • EXERCISES & SOLUTIONS • SIX PROJECTS

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